Exercises

Category Theory, M1 MPRI, ENS Paris-Saclay

First period, 2026-27

1 Preorders and monoids as categories

Exercise 1 (Preorders as thin categories)**.** A category C is *thin* if for any objects a, b of C, there is at most one morphism from a to b. A *preorder* is a set P equipped with a binary relation $\leq$ which is reflexive and transitive.

Prove that a small thin category is essentially the same thing as a preordered set. More precisely, define mutually inverse assignments between the collection of small thin categories and the collection of preorders.

Exercise 2 (Monoids as one-object categories)**.** Prove that a small category with one object is essentially the same thing as a monoid.

2 Isos, epis, monos

Definition A. Let $f\colon a \to b$ be a morphism in a category C. For any object c of C, we define a function $f_c^*\colon C(b,c) \to C(a,c)$ by $f_c^*(h) := hf$ for any $h\colon b \to c$. We also define a function $f_c^+\colon C(c,a) \to C(c,b)$ by $f_c^+(g) := fg$ for any $g\colon c \to a$.

Exercise 3 (Isomorphisms in a category)**.** Let $f\colon a \to b$ be a morphism in a category C. Prove that the following four properties are equivalent:

(i) f is an isomorphism, i.e., there exists $g\colon b \to a$ such that $fg = \mathrm{id}_b$ and $gf = \mathrm{id}_a$;

(ii) f has a left inverse and a right inverse, i.e., there exist $p\colon b \to a$ and $s\colon b \to a$ such that $fs = \mathrm{id}_b$ and $pf = \mathrm{id}_a$;

(iii) for any object c of C, the function f_c^* is a bijection;

(iv) for any object c of C, the function f_c^+ is a bijection.

Exercise 4 (Iso $\subsetneq$ Mono $\cap$ Epi)**.** This exercise shows that the class of isomorphisms is included in the class of morphisms that are both mono and epi, and that this inclusion is strict in general.

a. Prove that an isomorphism is both a monomorphism and an epimorphism.

b. Prove that, in the category **Mon** of monoids and homomorphisms, the inclusion function $i\colon \mathbb{N} \hookrightarrow \mathbb{Z}$ is both a monomorphism and an epimorphism, but not an isomorphism.

Hint. Recall that a monoid homomorphism must preserve any existing inverses.

Exercise 5 (Split epimorphisms)**.** Let C be a category. A morphism $p\colon a \to b$ in C is called a *split epimorphism* if it has a right inverse, i.e., if there exists $s\colon b \to a$ such that $ps = \mathrm{id}_b$.

a. Prove that a split epimorphism is an epimorphism.

b. Give an example of a category C in which not every epimorphism is a split epimorphism.

c. Prove that p is a split epimorphism if, and only if, for any object c of C, the function p_c^+ (see Definition A) is surjective.

d. Show, without using the Axiom of Choice, that a function is surjective if, and only if, it is an epimorphism in the category **Set**.

e. Recall that the Axiom of Choice (AC) states that, for any set $\mathcal{A}$ of non-empty sets, there exists a function f with domain $\mathcal{A}$ and such that, for every $A \in \mathcal{A}$, $f(A) \in A$. Prove that AC is equivalent to: in the category **Set**, every epimorphism is a split epimorphism.

3 Functors

Exercise 6. Let C be the category whose objects are the vector spaces $\mathbb{R}^d$ for each $d \in \mathbb{N}$, and whose morphisms are the linear transformations. Let $\mathbf{Mat}_{\mathbb{R}}$ be the category whose objects are natural numbers, whose morphisms from k to n are the $n \times k$ real matrices, and whose composition is given by matrix multiplication.

Define a functor F from $\mathbf{Mat}_{\mathbb{R}}$ to C which is an *isomorphism*, i.e., F is a bijection on objects and, for any objects n, k of $\mathbf{Mat}_{\mathbb{R}}$, the function $F\colon \mathbf{Mat}_R(k,n) \to C(Fk, Fn)$ is a bijection.

Exercise 7. Let Σ be a set. Using Exercise 2, regard the monoid Σ^* as a one-object category. Denote by $\mathbf{Set}_f$ the category of finite sets and functions between them. Prove that the collection of functors $F\colon \Sigma^* \to \mathbf{Set}_f$ is in bijection with the collection of pairs (Q, δ), where Q is a finite set and $\delta\colon \Sigma \times Q \to Q$ is a function.

Exercise 8. Let $F\colon C \to D$ be a functor.

a. Prove that F preserves isomorphisms, i.e., if f is an iso in C, then Ff is an iso in D.

b. Show that F does not necessarily preserve monomorphisms or epimorphisms.

Exercise 9. A *quiver* is a tuple $G = (V, E, d, c)$, where V and E are sets, and $d, c\colon E \rightrightarrows V$ are functions. For $a, b \in V$, a *path* from a to b in G, is a list $p = (a_n, f_n, a_{n-1}, f_{n-1}, \ldots, a_1, f_1, a_0)$, for some $n \geq 0$, with $a_i \in V$ and $f_i \in E$ for each $0 \leq i \leq n$, such that $a_0 = a$, $a_n = b$, and, for each $0 < i \leq n$, $d(f_i) = a_{i-1}$ and $c(f_i) = a_i$; we write $p\colon a \to b$. The *category of paths* on G, PG, has set of objects V and set of morphisms the paths in G.

a. Define an appropriate identity and composition on PG and show that PG is a category.

b. Let C be a category. A *morphism* from G to C is a pair of functions $F_0\colon V \to \mathrm{ob}\, C$ and $F_1\colon E \to \mathrm{mor}\, C$ such that, for all $f \in E$, $F_1 f$ has domain $F_0(df)$ and codomain $F_0(cf)$. Prove that, for any morphism (F_0, F_1) from G to C, there is a unique functor $\bar{F}\colon PG \to C$ such that $\bar{F}a = F_0 a$ for all $a \in V$ and $\bar{F}(ce, e, de) = F_1 e$ for all $e \in E$.

Remark. The category PG is also called the *free category* on G.

4 Binary products and terminal objects

Exercise 10. Let P be a preorder, seen as a category (see Exercise 1).

a. Let $t \in P$. Show that t is terminal if, and only if, t is maximal.

b. Let $a, b, c \in P$. Show that c is a product of a and b if, and only if, c is a greatest lower bound (infimum) of a and b.

Exercise 11. a. Give an example of sets A and B such that the projection $A \times B \to B$ is not surjective.

b. Conclude that product projections need not be epimorphisms.

c. Show that the unique map to a terminal object need not be an epimorphism.

Definition B. Let I be an index set, C a category, a an object, and $(f_i \colon a \to b_i)_{i \in I}$ a family of morphisms with domain a. We say that the family $(f_i)_{i \in I}$ is *jointly monomorphic* if, for any pair of morphisms $g, h \colon x \rightrightarrows a$, if $f_i \circ g = f_i \circ h$ for every $i \in I$, then $g = h$.

We say that a jointly monomorphic family $(f_i \colon a \to b)_{i \in I}$ is *extremal* if, for any family of morphisms $(f'_i \colon a' \to b_i)_{i \in I}$ and any epimorphism $e \colon a \to a'$, if $f_i = f'_i \circ e$ for every $i \in I$, then e is an isomorphism.

Exercise 12. Let C be a category, let b_1, b_2 be objects in C and let $p_1 \colon b_1 \times b_2 \to b_1$, $p_2 \colon b_1 \times b_2 \to b_2$ be a product.

a. Prove that the family (p_1, p_2) is jointly monomorphic and extremal.

b. Let $(f_i \colon a \to b_i)_{i=1,2}$ be a family of morphisms with domain a. Prove that the family is jointly monomorphic if, and only if, $\langle f_1, f_2 \rangle$ is a monomorphism.

5 Binary coproducts and initial objects

Definition C. Let C be a category and a_1, a_2 objects of C. A pair of morphisms $i_1 \colon a_1 \to b$ and $i_2 \colon a_2 \to b$ is a *coproduct* if it is a product in C^{op}. An object a of C is *initial* if it is terminal in C^{op}.

Exercise 13. a. Write down a more concrete definition of 'coproduct' and of 'initial'.

b. Explain why coproducts and initial objects are unique up to isomorphism.

c. Show that coproducts and an initial object exist in **Set**.

d. What are a coproduct and initial object in a preorder?

Notation. A coproduct of two objects a_1, a_2 is denoted $a_1 + a_2$, and an initial object is denoted 0. The copairing of two morphisms $f_1 \colon a_1 \to c$ and $f_2 \colon a_2 \to c$ is denoted $[f_1, f_2] \colon a_1 + a_2 \to c$. The functions $i_k \colon a_k \to a_1 + a_2$ are called *coprojections*, or also *injections* or *inclusions* (note that they are not necessarily monomorphisms, by the dual of Exercise 11).

Exercise 14. Let C be a category in which any two objects have a product and a coproduct.

a. Show that, for any objects a, b, c, there exists a morphism $(a \times b) + (a \times c) \to a \times (b + c)$.

b. Does there always exist an isomorphism?

Exercise 15. Consider the monoids $(\mathbb{N}, 0, +)$ and $(\{a, b\}^*, \epsilon, \cdot)$.

a. Prove that $\{a, b\}^*$ is a coproduct of $\mathbb{N}$ and $\mathbb{N}$ in the category **Mon**.

b. What is the coproduct of $\{a, b\}^*$ and $\mathbb{N}$ in **Mon**?

c. What is the coproduct of $\mathbb{Z}$ and $\mathbb{Z}$ in the category of groups?

Exercise 16. Let Σ be a finite set. Consider the following category, which we denote $\Sigma/\mathbf{Mon}$:

- objects are pairs (M, f), where M is a monoid, and $f\colon \Sigma \to M$ is a function from Σ to (the underlying set of) M;
- a morphism $(M, f) \to (N, g)$ is a monoid homomorphism $h\colon M \to N$ such that $h \circ f = g$.

Show that $\Sigma/\mathbf{Mon}$ has an initial object.

Exercise 17. Let C be a category with a terminal object 1 and an initial object 0. Prove that any morphism $f\colon 1 \to 0$ is an isomorphism.

6 Equalizers and coequalizers

Exercise 18. Prove that any equalizer is a monomorphism.

Exercise 19. Let $f, g\colon A \rightrightarrows B$ be two functions in **Set**. Define $\sim$ to be the smallest equivalence relation on B such that $f(a) \sim g(a)$ for every $a \in A$. Prove that the quotient map $e\colon B \to B/\!\sim$ is a coequalizer of f and g.

7 Pullbacks and pushouts

Definition D. The *walking span*, Λ, is the (poset) category with three objects and two non-identity morphisms as depicted on the left below. The *walking cospan*, V, is its opposite category, as depicted on the right below.



A *span* in a category C is a diagram of shape Λ, and a *cospan* is a diagram of shape V.

A *pushout*[1] is a colimit of a span, and a *pullback*[2] is a limit of a cospan.

Exercise 20. a. Give more explicit definitions of pushout and pullback.

[1] In French: *somme amalgamée*.

[2] In French: *produit fibré*.

b. Let $f_1 \colon b_1 \to a$ and $f_2 \colon b_2 \to a$ be morphisms in a category C. Suppose that a product $b_1 \times b_2$ of b_1 and b_2 exists in C, with projections $p_i \colon b_1 \times b_2 \to b_i$ for $i = 1, 2$, and that the pair of morphisms $f_1 p_1, f_2 p_2 \colon b_1 \times b_2 \rightrightarrows a$ has an equalizer, $e \colon c \to b_1 \times b_2$. Prove that $(p_1 e, p_2 e)$ is a pullback of (f_1, f_2).

c. Let $f_1 \colon B_1 \to A$ and $f_2 \colon B_2 \to A$ be a cospan in **Set**. Show that there exists a pullback with summit $P := \{(b_1, b_2) \in B_1 \times B_2 \mid f_1(b_1) = f_2(b_2)\}$.

d. Let $f_1 \colon A \to B_1$ and $f_2 \colon A \to B_2$ be a span in **Set**. Compute its pushout.

Exercise 21. Let C be a category containing the following two commuting squares.

$$\begin{array}{ccccc} a & \xrightarrow{p} & b & \xrightarrow{t} & e \\ \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle r} & & \downarrow{\scriptstyle v} \\ c & \xrightarrow{s} & d & \xrightarrow{u} & f \end{array}$$

Suppose that (r, t) is a pullback of (u, v). Prove that (tp, q) is a pullback of (v, us) if, and only if, (p, q) is a pullback of (r, s).

Exercise 22. Let C be a category in which pullbacks and a terminal object exist.

a. Prove that binary products exist in C.

b. Prove that all equalizers exist in C.

Exercise 23. Prove that a morphism $f \colon a \to b$ is mono if, and only if, the square $\begin{array}{ccc} a & \xrightarrow{\mathrm{id}_a} & a \\ \downarrow{\scriptstyle \mathrm{id}_a} & & \downarrow{\scriptstyle f} \\ a & \xrightarrow{f} & b \end{array}$ is a pullback.

8 Limits in general

Exercise 24. The aim of this exercise is to prove that *a category with small products and equalizers is complete.*

Let C be a category and let $D \colon I \to C$ be a diagram in C, with I small. For each morphism t in I, write dt for its domain and ct for its codomain.

Suppose that the family of objects $(D_i)_{i \in \mathrm{ob}\, I}$ has a product $(p_i \colon P \to D_i)_{i \in \mathrm{ob}\, I}$ in C, and that the family of objects $(D_{ct})_{t \in \mathrm{mor}\, I}$ has a product $(q_t \colon Q \to D_{ct})_{t \in \mathrm{mor}\, I}$ in C.

Consider the morphisms $f, g \colon P \rightrightarrows Q$ defined as

$$f := \langle p_{ct} \rangle_{t \in \mathrm{mor}\, I} \text{ and } g := \langle Dt \circ p_{dt} \rangle_{t \in \mathrm{mor}\, I} \ ,$$

that is, $q_t \circ f = p_{ct}$ and $q_t \circ g = Dt \circ p_{dt}$ for every $t \in \mathrm{mor}\, I$.

Suppose that $e \colon E \to P$ is an equalizer of the pair (f, g).

Prove that $(p_i \circ e \colon E \to D_i)_{i \in \mathrm{ob}\, I}$ is a limit cone for D.

Definition E. A poset $(I, \leq)$ is *down-directed* if I is non-empty and, for any $a, b \in I$, there exists $c \in I$ such that $c \leq a$ and $c \leq b$. A diagram $D \colon I \to C$ is *projective* if I is a down-directed poset. A category C *has projective limits* if any projective diagram in C has a limit.

Exercise 25. The aim of this exercise is to prove that *a category with finite products and projective limits has all small products.* Combined with the previous exercise, it follows that *a category with finite and projective limits is complete.*

Let $(a_j)_{j\in J}$ be a family of objects in a category C. Define I to be the set of finite subsets of J, ordered by reverse inclusion, i.e., for $S, T \in I$, $S \leq T$ if, and only if $T \subseteq S$.

a. Prove that $(I, \leq)$ is down-directed.

For each S in I, suppose that $(p_i^S \colon D_S \to a_i)_{i\in S}$ is a product of $(a_i)_{i\in S}$ in C.

b. Show that whenever $S, T \in I$ and $S \leq T$, there exists a unique morphism $D_{ST} \colon D_S \to D_T$ in C such that, for any $i \in T$, we have $p_i^T \circ D_{ST} = p_i^S$.

c. Conclude that $D \colon I \to C$ is a diagram.

Suppose that $(q_S \colon P \to D_S)_{S\in I}$ is a limit of D.

d. Show that $(p_j^{\{j\}} \circ q_{\{j\}} \colon P \to a_j)_{j\in J}$ is a product of the family $(a_j)_{j\in J}$.

Exercise 26. Let C be a small category in which all small families have products. Prove that C is thin and complete.

9 The Yoneda lemma

The exercises in this section lead to a proof of the Yoneda Lemma. We first examine two special cases: the case of a (totally ordered) thin category in Exercise 28 and the case of a monoid in Exercise 29. These exercises could be solved quickly by appealing to the result of Exercise 30.a, but it is instructive to also find direct proofs.

Recall from Definition A that, for a morphism $f \colon a \to b$ in a category C and c an object of C, we write $f_c^* \colon C(b, c) \to C(a, c)$ for the function defined by $f_c^*(h) := hf$ for any $h \colon b \to c$, and $f_c^+ \colon C(c, a) \to C(c, b)$ for the function defined by $f_c^+(g) := fg$ for any $g \colon c \to a$.

Definition F. Let C be a (locally small, as we always assume by default) category.

For any object c of C, we write $\mathbf{y}c$ for the functor $C^{\mathrm{op}} \to \mathbf{Set}$ which sends an object a of C to the set $(\mathbf{y}c)a := C(a, c)$, and a morphism $f \colon a \to b$ of C to the function $(\mathbf{y}c)f := f_c^* \colon C(b, c) \to C(a, c)$.

For any morphism $g \colon c \to c'$ of C, we write $\mathbf{y}g$ for the natural transformation $\mathbf{y}c \Rightarrow \mathbf{y}c'$ whose component at an object a of C is the function $(\mathbf{y}g)_a := g_a^+ \colon C(a, c) \to C(a, c')$.

In what follows, when C is a category, we write $\widehat{C}$ for the category whose objects are functors $C^{\mathrm{op}} \to \mathbf{Set}$, and whose morphisms are natural transformations between such functors.

Exercise 27. Let C be a category.

a. Prove that, for any object c of C, $\mathbf{y}c$ is a functor.

b. Prove that, for any morphism g of C, $\mathbf{y}g$ is a natural transformation.

c. Prove that $\mathbf{y} \colon C \to \widehat{C}$ is a functor.

Exercise 28 (Yoneda lemma for ω)**.** Let ω be the total order of natural numbers, viewed as a thin category.

a. Show that functors $F\colon \omega^{\mathrm{op}} \to \mathbf{Set}$ correspond bijectively to inverse sequences of sets

$$\cdots \longrightarrow X_{n+1} \xrightarrow{f_n} X_n \longrightarrow \quad \cdots \quad \longrightarrow X_2 \xrightarrow{f_1} X_1 \xrightarrow{f_0} X_0$$

i.e., the data of, for each $n \in \omega$, a set X_n, and a function $f_n\colon X_{n+1} \to X_n$.

b. Describe, for any $n \in \omega$, the sequence corresponding to the functor $\mathbf{y}n\colon \omega^{\mathrm{op}} \to \mathbf{Set}$.

c. Let $n \in \omega$ and let $(f_n\colon X_{n+1} \to X_n)_{n\in\omega}$ be an inverse sequence of sets with corresponding functor $F\colon \omega^{\mathrm{op}} \to \mathbf{Set}$. Prove that natural transformations $\mathbf{y}n \Rightarrow F$ are in bijective correspondence with elements of X_n.

Exercise 29 (Yoneda lemma for monoids)**.** Let M be a monoid, viewed as a category with one object, denoted $\star$.

a. Show that functors $F\colon M^{\mathrm{op}} \to \mathbf{Set}$ correspond bijectively to *actions* of M on a set X. Here, by an action of M on X we mean a function $\bullet\colon X \times M \to X$ such that, for any $x \in X$, $x \bullet 1_M = x$, and for any $m, n \in M$, $x \bullet (mn) = (x \bullet m) \bullet n$.

b. Describe the action corresponding to the functor $\mathbf{y}\star$.

c. Let $F\colon M^{\mathrm{op}} \to \mathbf{Set}$ be a functor and let $\bullet\colon X \times M \to X$ be the corresponding action. Give a bijection between X and the set of natural transformations $\mathbf{y}\star \Rightarrow F$.

Exercise 30 (The Yoneda Lemma in general)**.** Let C be a category. For any functor $F\colon C^{\mathrm{op}} \to \mathbf{Set}$ and any object c of C, we define a function

$$\begin{aligned} \mathsf{e}_{c,F}\colon \widehat{C}(\mathbf{y}c, F) &\to Fc \\ \alpha &\mapsto \alpha_c(\mathrm{id}_c)\ . \end{aligned}$$

a. Prove that $\mathsf{e}_{c,F}$ is a bijection by constructing an inverse function.

b. Prove that e is natural in F, i.e., for any morphism $\beta\colon F \Rightarrow G$ in $\widehat{C}$, the diagram

$$\begin{array}{ccc} \widehat{C}(\mathbf{y}c, F) & \xrightarrow{\mathsf{e}_{c,F}} & Fc \\ \downarrow{\scriptstyle \beta^{+}_{\mathbf{y}c}} & & \downarrow{\scriptstyle \beta_c} \\ \widehat{C}(\mathbf{y}c, G) & \xrightarrow{\mathsf{e}_{c,G}} & Gc \end{array}$$

commutes.

c. Prove that e is natural in c, i.e., for any morphism $g\colon c \to d$ in C, the diagram

$$\begin{array}{ccc} \widehat{C}(\mathbf{y}d, F) & \xrightarrow{\mathsf{e}_{d,F}} & Fd \\ \downarrow{\scriptstyle (\mathbf{y}g)^{*}_{F}} & & \downarrow{\scriptstyle Fg} \\ \widehat{C}(\mathbf{y}c, F) & \xrightarrow{\mathsf{e}_{c,F}} & Fc \end{array}$$

commutes.

Exercise 31. Prove that the functor $\mathbf{y}\colon C \to \widehat{C}$ is full and faithful.

10 Adjunctions

Exercise 32. Let P and Q be posets, viewed as thin categories. Let $f\colon P \to Q$ be a functor, which we recall is the same thing as a monotone function.

a. Write down what it means in this case that a functor $g\colon Q \to P$ is right adjoint to f.

b. Prove that, if f has a right adjoint, then f preserves any supremum that exists in P.

c. Let $i\colon \mathbb{Z} \hookrightarrow \mathbb{Q}$ be the inclusion function. Does it have a right adjoint? Does it have a left adjoint?

d. Same questions for the inclusion function $\mathbb{Q} \hookrightarrow \mathbb{R}$.

Exercise 33. Let $\mathbf{Set}_*$ be the category of *pointed sets*, whose objects are pairs (X, x), with $x \in X$, and whose morphisms $(X, x) \to (Y, y)$ are functions $f\colon X \to Y$ such that $f(x) = y$. Prove that the functor $U\colon \mathbf{Set}_* \to \mathbf{Set}$, which sends (X, x) to X and f to f, has a left adjoint.

Exercise 34. Prove that the forgetful functor $U\colon \mathbf{Mon} \to \mathbf{Set}$ has a left adjoint.

Exercise 35. Recall from Exercise 9 the definition of a category PG for any quiver G. (If you did not yet do Exercise 9, now is a good time.) A *morphism* from a quiver $G = (V_G, E_G, d_G, c_G)$ to a quiver $H = (V_H, E_H, d_H, c_H)$ is a pair of functions $f_0\colon V_G \to V_H$ and $f_1\colon E_G \to E_H$ such that, $d_H \circ f_1 = f_0 \circ d_G$ and $c_H \circ f_1 = f_0 \circ c_G$. This gives a category $\mathbf{Quiver}$ of quivers. Show that the forgetful functor $U\colon \mathbf{Cat} \to \mathbf{Quiver}$ from the category of small categories to the category of quivers has a left adjoint, which is given on objects by the construction $G \mapsto PG$.

Exercise 36. Let $F\colon C \leftrightarrows D : G$ be a pair of functors and, for each c in C and $d \in D$, let $\alpha_{c,d}\colon D(Fc, d) \xrightarrow{\cong} C(c, Gd)$ be a bijection. Prove that the following are equivalent:

(i) α is natural in both coordinates;

(ii) for any morphisms $f\colon Fc \to d$, $g\colon Fc' \to d'$, $h\colon c \to c'$, $k\colon d \to d'$, in (1), the diagram on the left commutes if, and only if, the diagram on the right commutes:

$$\begin{array}{ccc} Fc & \xrightarrow{f} & d \\ \downarrow{\scriptstyle Fh} & & \downarrow{\scriptstyle k} \\ Fc' & \xrightarrow{g} & d' \end{array} \qquad\qquad \begin{array}{ccc} c & \xrightarrow{f'} & Gd \\ \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle Gk} \\ c' & \xrightarrow{g'} & Gd' \end{array} \tag{1}$$

where, in the diagram on the right, $f' := \alpha_{c,d}(f)$ and $g' := \alpha_{c',d'}(g)$.

Exercise 37. A (bounded) *distributive lattice* is a poset L in which all finite suprema ($\vee$) and finite infima ($\wedge$) exist, and, for any $a, b, c \in L$, the equality $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ holds. A *morphism* from a distributive lattice L to a distributive lattice M is a function which preserves finite suprema and finite infima. This yields a category $\mathbf{DL}$. We write $\mathbf{DL}_{\text{fin}}$ for the full subcategory on the finite distributive lattices.

a. Prove that the forgetful functor $U\colon \mathbf{DL}_{\text{fin}} \to \mathbf{Set}_{\text{fin}}$ has a left adjoint.

Hint. For any finite set X, let FX be the lattice of upward closed subsets of the 'cube' poset $\{0,1\}^X$, equipped with an appropriate partial order, and define $\eta_X\colon X \to UFX$ by sending $p \in X$ to $\eta_X(p) := \{v \in \{0,1\}^X \mid v(p) = 1\}$.

Remark. This exercise implies that, for any $n \in \mathbb{N}$, the number $D(n) := |Fn|$ is finite. The number $D(n)$ is called the nth *Dedekind number*. The exact value of $D(9)$ was only computed in 2023, and the exact value of $D(10)$ is not known.

b. Does the extension $U\colon \mathbf{DL} \to \mathbf{Set}$ have a left adjoint?

11 Universal elements and arrows

In this section, let C be a category and $F\colon C^{\text{op}} \to \mathbf{Set}$ a functor.

Definition G. An *element* of the functor F is a pair (c, x), where c is an object of C and x is an element of the set Fc. Given two elements (c, x) and (d, y) of the functor F, a *morphism* from (c, x) to (d, y) is a morphism $f\colon c \to d$ in C such that $Ff(y) = x$. We write $\int F$ for the category of elements of the functor F.

Exercise 38. Prove that $\int F$ is a category.

Definition H. An element (c, x) of F is *universal* if the unique (by the Yoneda Lemma) natural transformation, $\alpha\colon \mathbf{y}c \Rightarrow F$, such that $\mathsf{e}_{c,F}(\alpha) = x$, is an isomorphism.

Exercise 39. Let (c, x) be an element of F. Show that (c, x) is universal if, and only if, (c, x) is a terminal object of $\int F$.

Exercise 40. This exercise shows how to turn around the arrows in the previous two exercises. Let D be a category and let $G\colon D \to \mathbf{Set}$ be a (covariant!) functor. Define $\int G$ to be the category whose objects are elements of G and whose morphisms $(c, x) \to (d, y)$ are the morphisms $f\colon c \to d$ in C such that $Gf(x) = y$. This category is also denoted $\int G$, although this notation can lead to confusion, see item a.

a. Let $C := D^{\text{op}}$ and let $G'\colon C^{\text{op}} \to \mathbf{Set}$ be the same functor as G, but viewed contravariantly. Show that the category $\int G$, as defined in this exercise, is the same as $(\int G')^{\text{op}}$, as defined in Definition G.

b. Write $\mathbf{y}'\colon D^{\text{op}} \to \mathbf{Set}^D$ for the Yoneda embedding of D^{op} and, for any object d of D, $\mathsf{e}'_{c,G}$ for the bijection $\mathbf{Set}^D(\mathbf{y}'c, G) \xrightarrow{\cong} Gc$. An element (c, x) of G is *universal* if the unique natural transformation $\alpha\colon \mathbf{y}'c \Rightarrow G$ with $\mathsf{e}'_{c,G}(\alpha) = x$ is an isomorphism. Prove that an element (c, x) of G is universal if, and only if, (c, x) is an initial object in $\int G$.

Conclusion: the meaning of 'universal element' depends on whether the functor is seen as covariant or contravariant.

Definition I. A *universal arrow* from an object c of C to a functor $U\colon D \to C$ is a pair (d, u), where d is an object of D and $u\colon c \to Ud$ is a morphism such that, for any morphism $f\colon c \to Ue$, there exists a unique morphism $f'\colon d \to e$ in D such that $Uf' \circ u = f$.

Exercise 41. Let $U\colon D \to C$ be a functor, let c be an object of C and consider the functor $H\colon D \to \mathbf{Set}$ given by composing U with $\mathbf{y}'c\colon C \to \mathbf{Set}$. Prove that a universal arrow is the same thing as a universal element of the covariant functor H.

12 Cartesian closed categories

Recall that a *cartesian closed category* is defined as a category C with finite products such that, for any object a of C, the functor $(-) \times a \colon C \to C$ has a right adjoint; we will write $(-)^a \colon C \to C$ for this right adjoint.[3]

Exercise 42. Let C be a category with finite products. Prove that C is cartesian closed if, and only if, for any objects a, b of C, there exist an object b^a in C and a morphism $\epsilon_{a,b} \colon b^a \times a \to b$ such that, for any morphism $f \colon c \times a \to b$, there exists a unique morphism $f' \colon c \to b^a$ such that $\epsilon_{a,b} \circ (f' \times \mathrm{id}_a) = f$.

Exercise 43. Let P be a partially ordered set. Characterize in order-theoretic terms what it means for the category associated to P to be cartesian closed.

Exercise 44 (Exam 2023)**.** Let C be a cartesian closed category in which finite coproducts exist. Construct, for any objects a, b, c, a morphism $c^a \times c^b \to c^{a+b}$.

Exercise 45. Let P be a poset. An *ω-chain* in P is a countable sequence $(p_i)_{i \in \omega}$ such that $p_i \leq p_{i+1}$ for every $i \in \omega$. The poset P is *ω-chain-complete* if any ω-chain has a supremum in P. A function $f \colon P \to Q$ between ω-chain-complete posets is a *morphism* if it preserves suprema of ω-chains. Prove that the category of ω-chain-complete posets is cartesian closed.

Remark. D. Scott used a variant of this cartesian closed category in 1972 to obtain a model of the untyped λ-calculus, by constructing a non-trivial ω-chain-complete U that is isomorphic to both $U \times U$ and U^U. For more information, see, e.g., Ch. 18 in J. Lambek, P. J. Scott, *Introduction to higher order categorical logic*, Cambridge University Press (1986).

Exercise 46. Let C be a small category. Recall that $\widehat{C}$ is the category of functors $C^{\mathrm{op}} \to \mathbf{Set}$, with natural transformations between them. Prove that $\widehat{C}$ is cartesian closed.

Hint. First assume the exponential G^F exists, and use the Yoneda lemma to rewrite $G^F(a)$ as the set of natural transformations from $\mathbf{y}a$ to G^F.

[3] In the literature, you may also find the notations $[a \to b]$ or $a \Rightarrow b$ instead of b^a.